

Nexus Project Architecture Update (Final Implementation Plan)

Project Architecture Decision

After technical evaluation and updated timeline availability, Team Nexus is moving forward with **User Wallet + UGF architecture** instead of backend-only transaction execution.

This architecture preserves blockchain transparency while still delivering gasless donations.

This is the final implementation architecture for the hackathon MVP.

Why We Changed the Previous Plan

Previous Emergency Plan (Rejected)

Backend-driven flow:

Frontend

→ Backend API

→ UGF SDK

→ Base Sepolia

→ Donation Contract

Problem:

All blockchain donations would appear from backend wallet.

Example:

Instead of:

user_1 donated

user_2 donated

user_3 donated

Blockchain would show:

backend wallet donated

backend wallet donated

backend wallet donated

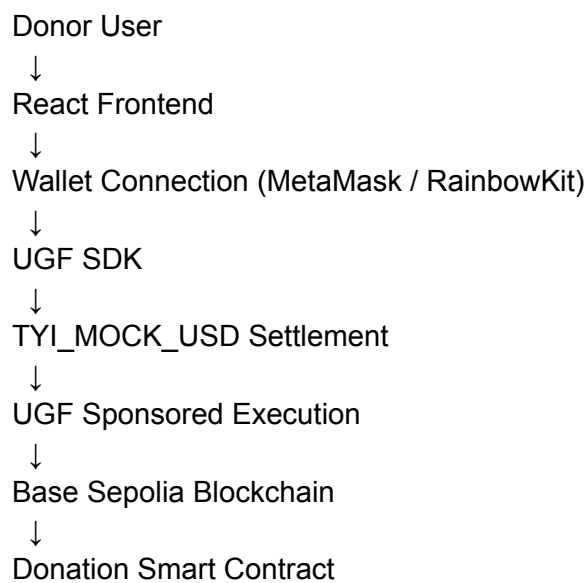
This breaks donor transparency.

Since our project's core value is trust + transparency, this architecture is not ideal.

Final Architecture (Approved)

User Wallet + UGF Gasless Execution

Final flow:



Core Benefits

This architecture gives:

- ✓ Real donor wallet identity on-chain
 - ✓ True donation transparency
 - ✓ No ETH required for donors
 - ✓ UGF hackathon alignment
 - ✓ Strong demo story
 - ✓ Proper blockchain audit trail
 - ✓ Better judge impression
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Project Technology Stack

Frontend

Purpose: User interface + wallet interaction

Tech:

- React.js
- Tailwind CSS
- React Router
- RainbowKit
- wagmi
- viem

Responsibilities:

- wallet connection
 - campaign listing
 - donation form
 - UGF integration trigger
 - transaction success display
 - dashboard UI
-

Blockchain

Purpose: Smart contract logic

Tech:

- Solidity
- Hardhat
- Base Sepolia
- OpenZeppelin

Responsibilities:

- campaign creation
 - donation recording
 - fund withdrawals
 - usage records
 - NGO verification
-

UGF Integration

Purpose: Gasless transaction execution

Tech:

- @tychilabs/ugf-testnet-js

Responsibilities:

- wallet authentication
 - transaction quote generation
 - TYI payment settlement
 - sponsored blockchain execution
-

Backend

Purpose: Application data management

Tech:

- Supabase
- Supabase PostgreSQL
- Supabase Storage

Responsibilities:

- NGO registration
- authentication
- campaign metadata
- usage documents
- receipts
- dashboard data

Backend DOES NOT execute donations.

Donation Flow

Step 1

User visits platform

Step 2

User connects wallet

Wallet:

- MetaMask
- or
- RainbowKit-supported wallet

Example wallet:

0xABC123...

Step 3

Frontend reads wallet address

This becomes donor identity.

Important:

This ensures blockchain mapping is correct.

Step 4

User views campaigns

Frontend fetches:

- active campaigns
 - goal
 - raised amount
 - NGO details
-

Step 5

User enters donation amount

Example:

50 TYI

Step 6

Frontend prepares smart contract transaction

Target contract:

`donateToCampaign(campaignId, amount, message)`

Transaction data encoded.

Step 7

Frontend authenticates with UGF

UGF SDK:

`client.auth.login()`

Uses connected wallet signer.

Step 8

UGF generates quote

UGF estimates:

- execution cost
 - TYI settlement amount
-

Step 9

UGF handles payment settlement

TYI MOCK_USD used for settlement.

No ETH needed.

Step 10

UGF sponsors blockchain execution

UGF executes transaction on Base Sepolia.

Step 11

Donation contract executes

Smart contract sees:

`msg.sender` = donor wallet

Example:

0xABC123...

NOT backend wallet.

Step 12

Blockchain updates

Contract stores:

- donor address
 - donation amount
 - campaign raised total
 - donation history
-

Step 13

Frontend receives tx hash

Example:

0x9abc...

Step 14

Frontend shows success

Display:

Donation Successful

[View on Base Explorer](#)

NGO Flow

NGO admins can:

- register/login
 - create campaign
 - define goal
 - upload campaign details
 - withdraw funds
 - upload expense proof
 - add usage records
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Transparency Dashboard

Public dashboard shows:

- campaign list
- total raised
- donation history
- donor addresses
- transaction explorer links
- usage logs
- receipts

This is our core product value.

Folder Architecture

Frontend

frontend/

└ src/


```
| ├── components/  
| ├── pages/  
| ├── routes/  
| ├── context/  
| ├── services/  
| ├── lib/  
| | ├── web3.jsx  
| | ├── ugf.js  
| | ├── contracts.js  
| | └── abi/
```

Blockchain

```
blockchain/  
├── contracts/  
|   └── Donation.sol  
├── ignition/  
├── scripts/  
├── test/  
└── hardhat.config.ts
```

Backend

```
Supabase  
├── auth  
├── database  
└── storage
```

Team Distribution

Member 1 — Blockchain

Responsibilities:

- Donation.sol final review
- TYI token integration
- Base Sepolia deployment
- ABI export
- contract testing

Deliverables:

- deployed contract

- contract address
 - ABI
-

Member 2 — UGF Integration

Responsibilities:

- UGF SDK integration
- wallet signer auth
- quote generation
- payment settlement
- sponsored execution
- test component

Files:

frontend/src/lib/ugf.js
frontend/src/components/UGFDonationTest.jsx

Deliverable:

successful gasless donation transaction

Member 3 — Frontend Donation UI

Responsibilities:

- campaign detail UI
 - donation form
 - wallet connect UI
 - success/failure states
 - explorer links
-

Member 4 — Backend / Supabase

Responsibilities:

- NGO auth
- database schema
- campaigns table
- fund usage records
- receipt storage

Member 5 — Dashboard + QA + Presentation

Responsibilities:

- transparency dashboard
 - donation history UI
 - bug testing
 - PPT
 - demo script
 - final submission
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Immediate Priority Order

Execution order:

Priority 1

Wallet connection stable

Priority 2

Deploy Donation contract

Priority 3

Implement real UGF SDK integration

MOST IMPORTANT

Need:

```
client.auth.login()
client.quote.get()
client.payment.x402.execute()
client.chains.evm.sponsorAndExecute()
```

Priority 4

Test successful gasless donation

Need:

real tx hash

Priority 5

Frontend donation flow

Priority 6

Dashboard

Priority 7

NGO admin flow

MVP Scope

Must Build:

- ✓ Wallet connect
- ✓ Campaign creation
- ✓ Gasless donation via UGF
- ✓ Donation smart contract
- ✓ Donation tracking
- ✓ Transparency dashboard
- ✓ NGO expense updates

Optional:

- QR donation
- NFT badges
- notifications
- fancy UI polish

Final Product Vision

Trustless disaster funding without crypto friction.

Users donate transparently on blockchain without needing ETH, while NGOs provide publicly verifiable fund usage updates.

Final Team Goal

Primary demo objective:

A fully working gasless blockchain donation platform using UGF + Base Sepolia + transparent donor tracking.